Do Now · retrieval

Dijkstra: shortest paths II

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What vertex does Dijkstra settle next?

Q2

What does a weighted edge record?

Q3

What does a predecessor in a path mean?

Q4

Distance to A=3, edge A–B=4, current distance B=10. Update B.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Track improvements — Relax each edge by comparing the candidate total with the current best-known distance.

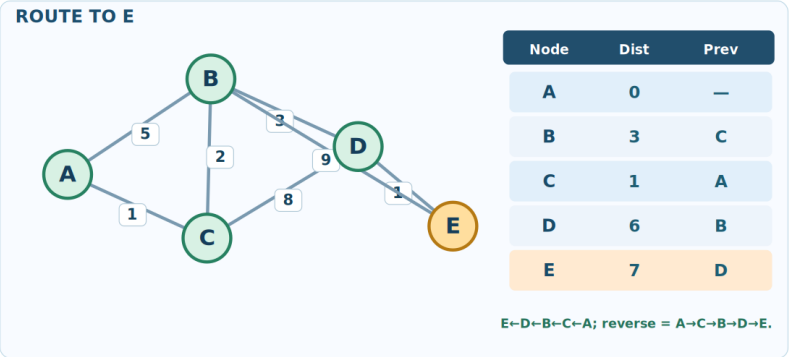
Core idea 2

Remember predecessors — When a distance improves, store the vertex from which the improvement came.

Core idea 3

Finish with the best distance — Here E reaches distance 7 after A, C, B and D have settled.

Key visual · complete concept



New Learning · applying the ideas

Core idea 4

Reconstruct backwards — Follow E←D←B←C←A, then reverse it to get the forward route.

Core idea 5

Handle unreachable vertices — A vertex remaining at infinity cannot be reached from this start along the graph's edges.

Vocabulary · use the correct technical terms

predecessor table — For each vertex, stores the prior vertex on its best-known path.

relaxation — Replace a distance when a cheaper route is discovered.

settled — Shortest distance is final for a vertex under non-negative edge weights.

infinity — Marker for an as-yet unreachable vertex.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

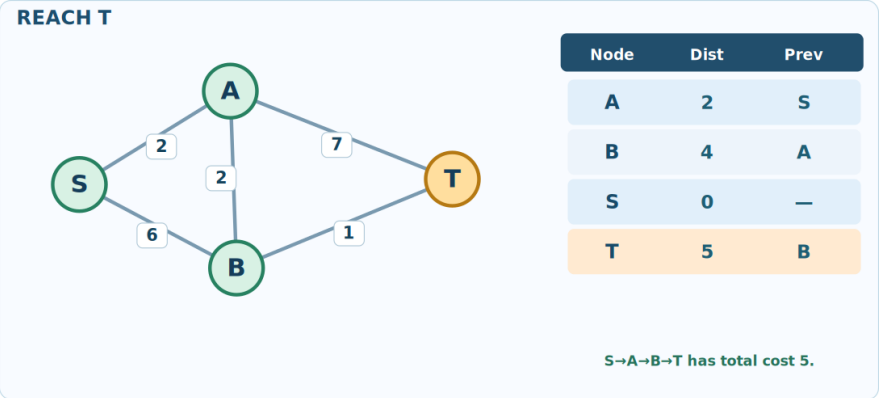
Problem

Find the shortest route S→T and reconstruct it.

Follow the worked method

1. Predict the next state — Start S=0: tentative A=2, B=6.
2. Update the table — Settle B=4; T improves to 5.
3. Update the table — Trace T←B←A←S, then reverse: S→A→B→T at cost 5.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict the next state
Start P=0 gives Q=4, R=1; settle R next.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Starting at U, edges U–V=4, U–W=1 and W–V=2. Find the shortest U→V path and total cost.

Question 2

Edges S–A=5, S–B=1, B–A=2, A–T=3 and B–T=9. Find S→T shortest distance, route and predecessor of T.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A graph contains a disconnected vertex Z. Explain what happens to tentative distance and predecessor for Z in Dijkstra, and when the algorithm may stop.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A predecessor table should be followed from the destination backwards before reversing.

Use key vocabulary from memory

1. Explain predecessor table in your own words.

2. Explain relaxation in your own words.

3. Explain settled in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.